



Exploring the Relationship Between Motion and Force

Viewpoint 1: Aristotle (fourth century BCE)

Only a continuous force can keep an object in motion.

Observation and
induction

Viewpoint 2: Galileo (early seventeenth century)

Motion does not require a force to be maintained.

Experimental
evidence and
reasoning

Question

Is there a common principle that accounts for the motion of all objects?

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Viewpoint 3: Newton (1687)

Force is the cause of changes in an object's state of motion.

Mathematical
synthesis

Section 5: Newton's First Law

1. Newton's First Law:

An object remains in its state of rest or uniform motion in a straight line unless a net external force causes it to change that state.

2. Inertia

The property by which an object maintains its original state of motion is called inertia.



Apply What You Have Learned



Water droplet in
free fall



Planet orbiting the Sun



Suitcase rolling
forward

Decoding Tianwen-1's Interplanetary Marathon

Background:

Tianwen-1, China's Mars probe, was launched from Earth and completed its Earth-to-Mars transfer in about 200–300 days. With only a limited amount of rocket fuel, the probe did not fly straight to Mars. Instead, it completed three key velocity changes during its journey.

Choose one of the following:

1. Draw a diagram explaining the orbital trajectory.
2. Write an inquiry report of about 500 words.
3. Give a 5-minute presentation.